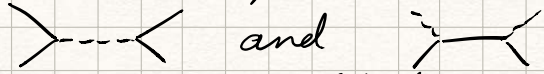


Summary of reading: EH 2.3, 2.6

§2.3 Examples from Yukawa Theory

We examined the tree level 4pt amplitudes in Yukawa theory:



Many useful properties of the spinor helicity variables are discovered in the context of these simple amplitudes:

- 1) Define spinor brackets: $\langle pq \rangle = \langle p | \cdot | q \rangle$ and $[Pq] = [P \cdot q]$
note the asymmetry of these brackets due to the ϵ -tensor: $\langle pq \rangle = -\langle qp \rangle$
 $\langle pq \rangle = [pq] = 0$
- 2) Converting Minkowski dot products to spinor helicity variables: $(p+q)^2 = \langle pq \rangle [pq]$
for any null momenta p and q .
- 3) $[Pq]^* = \langle pq \rangle$ for $p, q \in \mathbb{R}^{3,1}$
- 4) Incorporating γ -matrices: $\bar{u}_-(p) \gamma^\mu v_+(q) \equiv \langle p | \gamma^\mu | 2 \rangle$, $\bar{u}_+(p) \gamma^\mu v_-(q) \equiv [p | \gamma^\mu | q \rangle$
 $\bar{u}_\pm \gamma^\mu v_\pm = 0$
- 5) $\langle k | \gamma^\mu | p \rangle = \langle p | \gamma^\mu | k \rangle$ and $[k | \gamma^\mu | p \rangle^* = [p | \gamma^\mu | k \rangle$ (for $p, q \in \mathbb{R}^{3,1}$)
- 6) Fiertz identities: $\langle 1 | \gamma^\mu | 2 \rangle \langle 3 | \gamma_\mu | 4 \rangle = 2 \langle 13 \rangle [24]$
- 7) Momentum conservation in spinor helicity variables: $\sum_{i=1}^n |i\rangle [i] = 0$
- 8) Shouten identity (follows trivially from the fact that 3 vectors in a plane cannot be linearly independent)
 $|i\rangle \langle jk| + |j\rangle \langle ki| + |k\rangle \langle ij| = 0$

§2.6 Little Group scaling

little group = all $\Lambda_p \in \text{SO}(d-1, 1)$ that do not change the direction of the momentum p^μ . The angle and square brackets satisfy the massless Weyl equation $p|p\rangle = 0 = p|p]$ for $p^2 = 0$. The Weyl eq is invariant under the scaling: $|p\rangle \rightarrow t|p\rangle$ and $|p] \rightarrow t^{-1}|p]$ and is exactly the little group transformation in spinor helicity notation. (t is a phase for $p \in \mathbb{R}^{d-1,1}$ and $t \in \mathbb{C}$ for $p \in \mathbb{C}^{d-1,1}$)

only external particles of a Feynman diagram scale under the little group:

- 1) scalars: no scaling
- 2) fermions: spinors scale as t^{-2h} where $h = \pm \frac{1}{2}$
- 3) bosons: polarization vectors scale as t^{-2h} for $h = \pm 1$

$\Rightarrow$ for an amplitude of only massless particles, the little group scaling is

$$A_n(\{i_1, i_2, \dots, i_{2i}\}, \{j_1, j_2, \dots, j_{2j}\}) = t_i^{-2i} A_n(\{i_1, i_2, \dots, i_{2i}\}, \{j_1, j_2, \dots, j_{2j}\})$$

this is a powerful constraint that will be exploited many times. For example, the little group scaling uniquely fixes the 3pt amplitude up to an overall constant

$$A_3(1^{h_1} 2^{h_2} 3^{h_3}) = c \langle 12 \rangle^{h_3-h_2-h_1} \langle 13 \rangle^{h_2-h_1-h_3} \langle 23 \rangle^{h_1-h_2-h_3}$$

where we have used the fact that the 3pt amplitude of massless particles depends only on angle or square brackets.

on-shell

if we avoid making the assumption that 3pt amplitudes only depend on square or angle brackets we could determine the correct structure using dimensional analysis. For example, consider 3 gluon amplitude:

$$A_3(1^- 2^- 3^+) = g \frac{\langle 12 \rangle^3}{\langle 13 \rangle \langle 23 \rangle}$$

dimension -1
compatible to AA A

$$\text{or } A_3(1^- 2^- 3^+) = g' \frac{[13][23]}{[12]^3}$$

dimension -1

would need AA $\frac{3}{2}A$

cannot appear in local L.

note that $[g] = 0$ and $[g'] = 2$ since both amplitudes must have same dim

$$\text{in general: } \dim(A_n) = d - \frac{(d-2)}{2}n$$

Exercises! 2.1, 2.3, 2.4, 2.8, 2.32, 2.34 of EH and 33.2 of Srednicki

EH 2.1

$$p^\mu = (E, E \sin \theta \cos \varphi, E \sin \theta \sin \varphi, E \cos \theta)$$

$$p_{ab} = p_\mu (\sigma^\mu)_{ab} = \begin{pmatrix} -p^0 + p^3 & p^1 - ip^2 \\ p^1 + ip^2 & -p^0 - p^3 \end{pmatrix} = 2E \begin{pmatrix} \sin^2 \theta/2 & e^{-i\varphi} \cos \theta/2 \sin \theta/2 \\ e^{i\varphi} \cos \theta/2 \sin \theta/2 & -\cos^2 \theta/2 \end{pmatrix}$$

$$p^{\dot{a}b} = p_\mu (\bar{\sigma}^\mu)^{\dot{a}b} = -2E \begin{pmatrix} \cos^2 \theta/2 & e^{-i\varphi} \cos \theta/2 \sin \theta/2 \\ e^{i\varphi} \cos \theta/2 \sin \theta/2 & \sin^2 \theta/2 \end{pmatrix}$$

we are given

$$|p\rangle^{\dot{a}} = \sqrt{2E} \begin{pmatrix} \cos \theta/2 \\ \sin \theta/2 e^{i\varphi} \end{pmatrix}$$

and want to check that it satisfies $p_{ab} |p\rangle^{\dot{b}} = 0$. This is easily verified by using Mathematica to compute the matrix product.

Furthermore it is simple to verify (on Mathematica) that

$$\langle p|_{\dot{a}} = \sqrt{2E} \begin{pmatrix} -\sin \theta/2 e^{-i\varphi} & \cos \theta/2 \end{pmatrix}$$

$$[p]_{\dot{a}} = \sqrt{2E} \begin{pmatrix} \cos \theta/2 & \sin \theta/2 e^{-i\varphi} \end{pmatrix}$$

$$|p]_{\dot{a}} = \sqrt{2E} \begin{pmatrix} -\sin \theta/2 e^{-i\varphi} & \cos \theta/2 \end{pmatrix}^T$$

satisfy the Weyl equations:

$$\langle p|_{\dot{a}} p^{\dot{a}b} = 0$$

$$[p]_{\dot{a}} p^{\dot{a}b} = 0$$

$$p^{\dot{a}b} |p]_{\dot{b}} = 0$$

and the completeness relations:

$$p_{ab} = -|p]_{\dot{a}} \langle p|_{\dot{b}}$$

$$p^{\dot{a}b} = -|p\rangle^{\dot{a}} [p]_{\dot{b}}$$

EH 2.3 Prove the Fey identity $\langle 1 | \gamma^\mu | 2 \rangle \langle 3 | \gamma_\mu | 4 \rangle = 2 \langle 13 \rangle \langle 24 \rangle$

$$\begin{aligned}
 \langle 1 | \gamma^\mu | 2 \rangle \langle 3 | \gamma_\mu | 4 \rangle &= \bar{u}_-(p_1) \gamma^\mu v_+(p_2) \bar{u}_-(p_3) \gamma_\mu v_+(p_4) \\
 &= (0, \langle 1 | \dot{\alpha}) \begin{pmatrix} 0 & (\sigma^\mu)_{\dot{\alpha} b} \\ (\bar{\sigma}^\mu)^{\dot{\alpha} b} & 0 \end{pmatrix} \begin{pmatrix} | 2 \rangle_b \\ 0 \end{pmatrix} \\
 &\quad \times (0, \langle 3 | \dot{c}) \begin{pmatrix} 0 & (\sigma^\mu)_{\dot{c} d} \\ (\bar{\sigma}^\mu)^{\dot{c} d} & 0 \end{pmatrix} \begin{pmatrix} | 4 \rangle_d \\ 0 \end{pmatrix} \\
 &= (0, \langle 1 | \dot{\alpha}) \begin{pmatrix} 0 & (\sigma^\mu)_{\dot{\alpha} b} \\ (\bar{\sigma}^\mu)^{\dot{\alpha} b} & | 2 \rangle_b \end{pmatrix} (0, \langle 3 | \dot{c}) \begin{pmatrix} 0 & (\sigma^\mu)_{\dot{c} d} \\ (\bar{\sigma}^\mu)^{\dot{c} d} & | 4 \rangle_d \end{pmatrix} \\
 &= \langle 1 | \dot{\alpha} \langle 3 | \dot{c} \underbrace{(\bar{\sigma}^\mu)^{\dot{\alpha} b} (\bar{\sigma}_\mu)^{\dot{c} d}}_{-2 \epsilon^{\dot{\alpha} \dot{c}} \epsilon^{bd}} | 2 \rangle_b | 4 \rangle_d \quad (\text{from appendix}) \\
 &= -2 \langle 1 | \dot{\alpha} | 3 \rangle^{\dot{\alpha}} [4]^b | 2 \rangle_b \\
 &= +2 \langle 13 \rangle [24]
 \end{aligned}$$

EH 2.4 Show $\langle k | \gamma^\mu | k \rangle = 2k^\mu$ and $\langle k | P | k \rangle = 2P_0 k$

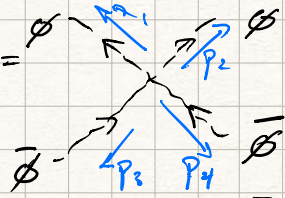
$$\begin{aligned}
 \langle k | \gamma^\mu | k \rangle &= \text{Tr} [\gamma^\mu | k \rangle \langle k |] \\
 &= \frac{1}{2} \text{Tr} [\gamma^\mu (| k \rangle \langle k | + | k \rangle \langle k |)] \\
 &= -\frac{1}{2} \text{Tr} [\gamma^\mu k] \\
 &= -\frac{1}{2} k_\nu (-4 \gamma^{\mu\nu}) \\
 &= 2k^\mu
 \end{aligned}$$

where we have used the fact that $[k | \gamma^\mu | p \rangle]^* = [p | \gamma^\mu | k \rangle$ for $p, k \in \mathbb{R}^3$
 $\langle k | P | k \rangle = 2P_0 k$ follows trivially $\square$

$$\begin{aligned}
 \langle k | \gamma^\mu | k \rangle &= \bar{u}_- \gamma^\mu v_+ = \bar{u}_- \gamma^\mu u_- \\
 (\langle k | \gamma^\mu | k \rangle)^\dagger &= \frac{v_+^\dagger}{\bar{v}_+} \gamma^{\mu\dagger} \frac{\bar{u}_-^\dagger}{u_-} \quad \frac{u_\pm}{\bar{u}_\pm} = \frac{v_\mp}{\bar{v}_\mp} \\
 &= \frac{v_+^\dagger}{\bar{v}_+} \gamma^0 \gamma^{\mu\dagger} \gamma^0 \frac{\bar{u}_-^\dagger}{u_-} \\
 &\Rightarrow (\langle k | \gamma^\mu | k \rangle)^\dagger = \langle k | \gamma^\mu | k \rangle \in \mathbb{R} \\
 \langle k | \gamma^\mu | p \rangle &= \bar{u}_-(k) \gamma^\mu v_+(p) \\
 \langle k | \gamma^\mu | p \rangle^\dagger &= \frac{v_+^\dagger(p)}{\bar{v}_+(p)} \gamma^{\mu\dagger} \frac{\bar{u}_-^\dagger(k)}{u_-(k)} \\
 &= \frac{v_+^\dagger(p)}{\bar{v}_+(p)} \gamma^\mu u_-(k) \\
 &= \bar{u}_-(p) \gamma^\mu v_+(k) \\
 &= [p | \gamma^\mu | k \rangle \\
 &\Rightarrow \langle k | \gamma^\mu | k \rangle = [k | \gamma^\mu | k \rangle
 \end{aligned}$$

EH 2.8 $\mathcal{L} = \bar{\psi} \gamma^\mu \partial_\mu \psi - \partial_\mu \bar{\psi} \partial_\mu \psi + \frac{1}{2} g \phi \bar{\psi} \psi + \frac{1}{2} g^* \bar{\phi} \psi^\dagger \psi^\dagger - \frac{1}{4} \lambda |\phi|^2$
 where ψ is a Weyl fermion and ϕ is a complex scalar.

$$i A_4(\phi \phi \bar{\psi} \bar{\psi}) = \phi \text{ --- } \phi = -i\lambda$$



$$i A_4(\phi f^- f^+ \bar{\phi}) =$$

(there is no $(1 \leftrightarrow 4)$ diagram
 b/c ϕ is a complex scalar)

$$\begin{aligned} &= \frac{ig^* - i\bar{u}_3(\not{p}_1 - \not{p}_2)u_2}{(\not{p}_1 + \not{p}_2)^2} ig \\ &= i|g|^2 \frac{\bar{u}_3(\not{p}_3 + \not{p}_4)u_2}{(\not{p}_3 + \not{p}_4)^2} \\ &= i|g|^2 \frac{[3](13)\langle 31+3 \rangle [3+4]\langle 4 \rangle \bar{4} \langle 4 \rangle [4] 2}{\langle 34 \rangle [34]} \\ &= i|g|^2 \frac{[34] \langle 42 \rangle}{\langle 34 \rangle [34]} \\ &= -i|g|^2 \frac{\langle 24 \rangle}{\langle 34 \rangle} \end{aligned}$$

$$i A_4(f^- f^- f^+ f^+) =$$

$$\begin{aligned} &= \frac{(ig)(ig^*)(-i)}{(\not{p}_3 + \not{p}_4)^2} \bar{u}_1 u_2 \bar{u}_4 u_3 \\ &= \frac{i|g|^2}{\langle 34 \rangle [34]} \langle 12 \rangle [43] \\ &= -i|g|^2 \frac{\langle 12 \rangle}{\langle 34 \rangle} \end{aligned}$$

EH 2.32

$$a) A_5 = g_a \frac{[13]^4}{[12][23][34][45][51]} \xrightarrow{\text{little group transformation}} t_1^{-2} t_2^2 t_3^{-2} t_4^2 t_5^2 A_5$$

$$\Rightarrow (h_1 h_2 h_3 h_4 h_5) = (1, -1, 1, -1, -1)$$

The mass dim of the amplitude is $[A_5] = 4 - 5 = -1 \Rightarrow [g_a] = 0$

$\Rightarrow$ Yang mills theory.

$$b) A_4 = g_b \frac{\langle 14 \rangle \langle 24 \rangle^2}{\langle 12 \rangle \langle 23 \rangle \langle 34 \rangle} \longrightarrow t_1^0 t_2^0 t_3^{-2} t_4^2 A_4$$

$$\Rightarrow (h_1 h_2 h_3 h_4) = (0, 0, 1, -1)$$

$$[A_4] = 0 \Rightarrow [g_b] = 0$$

we need 4 fields in the interaction term such that it has dim 4 so that $[g_b] = 0$.

scalar QED/YM

$$c) A_4 = g_c \frac{\langle 12 \rangle^2 [12]}{\langle 13 \rangle \langle 14 \rangle \langle 23 \rangle \langle 24 \rangle \langle 34 \rangle^2} \longrightarrow t_1^4 t_2^4 t_3^{-4} t_4^{-4} A_4$$

$$\Rightarrow (h_1 h_2 h_3 h_4) = (-2, -2, 2, 2)$$

$$[A_4] = 0 \Rightarrow [g_c] = -2$$

looks like a 4-graviton interaction: $R \sim (g_{\mu\nu})^2$ so we expect such an interaction from R^2

EH 2.32

define amplitude as: $A_3(1^{h_1} 2^{h_2} 3^{h_3})$ for $h_1, h_2 = \pm \frac{1}{2}$ and $h_3 = \pm 1$

under little group scaling. $A_3 \rightarrow t_1^{-2h_1} t_2^{-2h_2} t_3^{-2h_3} A_3$

$$\langle ij \rangle \rightarrow t_i t_j \langle ij \rangle \quad [ij] \rightarrow t_i^{-1} t_j^{-1} [ij]$$

while we have not covered it in our reading yet, 3-particle kinematics is special. For 3 massless particles,

$$\begin{aligned} \& \quad [12] \propto [23] \propto [13] \\ & \quad |1\rangle \propto |2\rangle \propto |3\rangle \end{aligned}$$

this implies:

- 1) amplitudes can only depend on either $| \rangle$ or $[]$ brackets, not both.
- 2) 3-particle massless amplitudes are non-zero only for complex momentum.

Thus, the 3pt amplitude is completely fixed to be

$$A_3^{h_1 h_2 h_3} = g \langle 12 \rangle^{h_3 - h_2 - h_1} \langle 13 \rangle^{h_2 - h_1 - h_3} \langle 23 \rangle^{h_1 - h_2 - h_3}$$

there are 3 classes of amplitudes we can have for a gluino-gluino-gluon amplitude:

$$1) (h_1, h_2, h_3) = (-\frac{1}{2}, -\frac{1}{2}, +1)$$

$$A_3^{\frac{1}{2}, -\frac{1}{2}, +1} = g \frac{\langle 12 \rangle^2}{\langle 13 \rangle \langle 23 \rangle} \quad \text{where } [g] = 1$$

$$2) (h_1, h_2, h_3) = (+\frac{1}{2}, -\frac{1}{2}, -1)$$

$$A_3^{+\frac{1}{2}, -\frac{1}{2}, -1} = g \frac{\langle 23 \rangle^2}{\langle 12 \rangle} \quad \text{where } [g] = 0$$

$$3) (h_1, h_2, h_3) = (-\frac{1}{2}, -\frac{1}{2}, -1)$$

$$A_3^{\frac{1}{2}, -\frac{1}{2}, -1} = g \langle 13 \rangle \langle 23 \rangle \quad \text{where } [g] = -1$$

what kind of Lagrangian could give us such amplitudes?

2) $\bar{\psi} A \psi$ is a natural candidate. Looking through §2.4 we see that A_3^{+-} has the same form as the QED amplitudes (which we know follow from $\mathcal{L}_{int} = \bar{\psi} A \psi$)

$$1) [g] = 1, [A] = \frac{3}{2}, [A] = 1 \Rightarrow g \int d^4x (\bar{\psi} \not{A} \psi) \quad (\text{non-local})$$

$$2) [g] = -1, [A] = \frac{3}{2}, [A] = 1 \Rightarrow g \bar{\psi} \not{A}^2 \psi$$

EH 2 33

From $A_3^{h_1 h_2 h_3} = g \langle 12 \rangle^{h_2 - h_1 - h_3} \langle 13 \rangle^{h_2 - h_3 - h_1} \langle 23 \rangle^{h_1 - h_2 - h_3}$ we find

$$A_3^{-2, -2, -2} = g \langle 12 \rangle^2 \langle 13 \rangle^2 \langle 23 \rangle^2 \quad \text{for } [g] = -5$$

$$A_3^{-2, -2, 2} = g \frac{\langle 12 \rangle^6}{\langle 13 \rangle^2 \langle 23 \rangle^2} \quad \text{for } [g] = -1$$

where $[A_3^{h_1 h_2 h_3}] = 4 - 3 = 1$

Comparing to the gluon amplitudes

$$A_3^{-1, -1, -1} = g \langle 12 \rangle \langle 13 \rangle \langle 23 \rangle \quad \text{for } [g] = -2$$

$$A_3^{-1, -1, 1} = g \frac{\langle 12 \rangle^3}{\langle 13 \rangle \langle 23 \rangle} \quad \text{for } [g] = 0$$

We see that the kinematic part of the gravity amplitudes is the square of kinematic part of the gluon amplitude $\rightarrow$ BCJ & color kin duality!

Sredneci 332

$$[\mathcal{J}_i, \mathcal{J}_j] = i \varepsilon_{ijk} \mathcal{J}_k$$

$$[\mathcal{J}_i, k_j] = i \varepsilon_{ijk} k_k$$

$$[k_i, k_j] = i \varepsilon_{ijk} \mathcal{J}_k$$

$$N_i = \frac{1}{2} (\mathcal{J}_i - i k_i)$$

$$N_i^\dagger = \frac{1}{2} (\mathcal{J}_i + i k_i)$$

$$\begin{aligned} [N_i, N_j] &= \frac{1}{4} [\mathcal{J}_i - i k_i, \mathcal{J}_j - i k_j] \\ &= \frac{1}{4} ([\mathcal{J}_i, \mathcal{J}_j] - i [\mathcal{J}_i, k_j] - i [k_i, \mathcal{J}_j] - [k_i, k_j]) \\ &= \frac{1}{4} (i \varepsilon_{ijk} \mathcal{J}_k - i i \varepsilon_{ijk} k_k + i i \varepsilon_{jik} k_k + i \varepsilon_{ijk} \mathcal{J}_k) \\ &= \frac{1}{2} \varepsilon_{ijk} (\mathcal{J}_k - i k_k) \\ &= i \varepsilon_{ijk} N_k \end{aligned}$$

$$\begin{aligned} [N_i^\dagger, N_j^\dagger] &= \frac{1}{4} ([\mathcal{J}_i, \mathcal{J}_j] + i [\mathcal{J}_i, k_j] + i [k_i, \mathcal{J}_j] - [k_i, k_j]) \\ &= \frac{1}{4} 2i \varepsilon_{ijk} (\mathcal{J}_k + i k_k) \\ &= i \varepsilon_{ijk} N_k^\dagger \end{aligned}$$

$$[N_i, N_j^\dagger] = \frac{1}{4} (\underbrace{[\mathcal{J}_i, \mathcal{J}_j]}_{i \varepsilon_{ijk} \mathcal{J}_k} + i \underbrace{[\mathcal{J}_i, k_j]}_{i^2 \varepsilon_{ijk} k_k} - i \underbrace{[k_i, \mathcal{J}_j]}_{(-i)^2 \varepsilon_{jik} k_k} + \underbrace{[k_i, k_j]}_{-i \varepsilon_{ijk} \mathcal{J}_k}) = 0$$

$$\Rightarrow \text{Lorentz group} = \text{SU}(2)_R \otimes \text{SU}(2)_L$$

